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FULL LENGTH ARTICLE

Endo-atmospheric maneuver penetration strategy based on generative adversarial reinforcement learning



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Abstract An intelligent endo-atmospheric penetration strategy based on generative adversarial reinforcement learning is proposed in this manuscript. Firstly, attack and defense adversarial models are established, and missile maneuver penetration problem is transformed into an optimal control problem, considering penetration, handover position and mid-terminal guidance velocity constraints. Then, Radau Pseudospectral method is adopted to generate data samples considering random perturbations. Furthermore, Generative Adversarial Imitation Learning Combined with Deep Deterministic Policy Gradient method (GAIL-DDPG) is designed, with internal process reward signals constructed to tackle long-term sparse reward in missile manuver penetration problem. Finally, penetration strategy is trained and verified. Simulation shows that using generative adversarial reinforcement learning, with sample library to learn expert experience in training early stage, the proposed method can quickly converge. Also, performance is further optimized with reinforcement learning exploration strategy in the later stage of training. Simulation shows that the proposed method has better engineering application ability compared with traditional reinforcement learning method.

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1. Introduction

With development of artificial intelligence and advancement of weapon technology, adversarial attack and defense is with increasingly significance in military operation. How to transform information advantages into decision-making advantages and then into adversarial advantages has become focus of the attack and defense sides in rapidly changing battlefield. Therefore, intelligent decision-making has become a research hotspot to solve the problem of attack and defense adversarial

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games, and attack missile penetration research in the novel battlefield is of great significance.

The intelligent game trajectory planning problem of hypersonic vehicle is essentially a combination of territorial defense problem[1] and pursuit and escape problem.[2] Due to unknown characteristics of interceptor missile maneuver capability and interception strategy, it is traditionally difficult for hypersonic vehicle to calculate optimal timing, strength and duration of ballistic penetration maneuver in actual flight process,[3] which brings practical difficulties to the application of penetration maneuver strategy. To solve this problem, a novel penetration method is proposed in this manuscript. The proposed method makes full use of hypersonic vehicle detection capability, perceives interception threat in real time in the process of penetration interception and confrontation, quickly generates a countermeasure strategy based on its flight state, to solve endo-atmospheric penetration problem. Furthermore, the proposed method also consider target strike needs, thus has strong engineering application value.

Current solution to interception penetration of hypersonic vehicle in complex battle environment can mainly be divided into trajectory planning and maneuver confrontation according to classification of penetration method. By modeling various threat zones (detection radar, artillery range, mountain terrain), and calculating threat level functions, the trajectory planning problem can be formulated into an optimal control problem, which require minimizing the path threat and energy consumption. Then the problem can be solved through geometric dynamics waypoint tracking method,[4] improved dynamic programming method,[5] Gaussian Pseudospectral method,[6] etc. However, detour trajectory planning is a relatively conservative algorithm, there may exist threats that cannot be avoided or the combat tasks cannot be achieved afterwards. In the maneuver confrontation problem, Wang et al.[7] designed a one-to-one optimal guidance law capable of penetration interception and target attacking. Wang et al.[8] proposed a Search-Resampling-Optimization (SRO) framework to solve the low computational efficiency of optimal control method under scenarios with complicated obstacles by paving a series of safe dispatch corridors. Wang et al.[9] proposed a multi-missile optimal cooperative decoy penetration guidance law based on induced collision strategy by inducing interception missiles from different targets to collide with each other. Liu et al.[10] used differential game guidance law to achieve multiple vehicles evading interception and simultaneously striking targets from different attack directions. Generally, these methods suffer from problems such as complex differential equations, multiple state variables, nonlinear or coupled constraints in optimal control or differential game theory solution to optimal maneuvering strategy of attacking vehicle and interceptors in the adversarial process, which pose great difficulty in seeking analytical solutions. In addition, it is generally necessary to obtain position, velocity, and acceleration information of both sides in the attack and defense confrontation to establish accurate models of attack and defense, which has poor adaptability when applied in battlefields with complex unknown disturbances.

The development of swarm intelligence theory[11] provides a novel research direction for the design of penetration guidance laws. The swarm intelligence methods[12–14] can be divided into two categories. One is cluster intelligence represented by biological heuristic algorithms, and the other is machine intelligence represented by multi-agent reinforcement learning. The former includes representative algorithms such as particle swarm optimization,[15] ant colony optimization algorithm,[16] artificial bee colony algorithm,[17] etc. However, biological heuristic swarm intelligence algorithm is sensitive to initial value of the optimization problem and is prone to getting stuck in local optima. Moreover, as the optimization problem progresses, the convergence rate of the optimal solution normally declines.

An improved multi-objective genetic algorithm[18] is proposed and an efficient unlocking method is constructed to address the unavoidable dead-lock phenomenon meanwhile maintaining the population randomness. The latter is deep reinforcement learning method, in which intelligent agent can iteratively optimize the penetration strategy of attacking vehicle by constantly interacting with battlefield environment for trial and error. It is a typical representative of swarm intelligence methods. Since deep reinforcement learning methods do not rely on models and have good generalization ability, they are suitable for solving difficult modeling and endo-atmospheric complex sequential decision-making problems. Air combat decision-making based on deep reinforcement learning in recent years are comprehensively reviewed, and the research direction of applying multi-agent decision-making in more complex air combat environments in the future is pointed out by researchers in Ref. 19. An endo-atmospheric sequential penetration method is proposed based on the Repetitive Batch Training Deep Deterministic Policy Gradient (RBT-DDPG)deep reinforcement learning method.[20] Out of sight air combat countermeasures are achieved using Long Short-Term Memory Deep Q-Network(LSTM-DQN) deep reinforcement learning method.[21] Although deep reinforcement learning method has good generalization ability, solely relies on environment exploration to gain experience, and improving strategies based on experience replay, thus suffers from long training time and slow convergence rate problems.[22–23] To tackle this problem, Ref. 24 proposed a penetration and strike guidance law for fighter jets based on optimal control and generative adversarial imitation learning method, and achieved good results. Through knowledge construction and accumulation, training of deep reinforcement learning can be hot started, fully leveraging the capabilities by using knowledge and data collaboration. It can also accelerate network convergence, and improve instruction accuracy at the same time.

Endo-atmospheric sequential decision-making problem of continuous maneuvering and countermeasure of attack hypersonic vehicles are studied in this manuscript. Firstly, the endo-atmospheric integrated penetration and guidance problem is formulated utilizing optimal control theory. The optimal control problem is then solved using Gaussian Pseudospectral method, obtaining optimal guidance law. Subsequently, dataset was constructed, with random disturbances and error models established to obtain battlefield environment close to actual flight. The dataset was then solved and trained using generative adversarial imitation learning. Finally, DDPG algorithm

was combined to improve penetration strategy network, resulting in a faster convergence rate and better online penetration guidance strategy performance.

The main contributions of this manuscript are as follows:

(1) An intelligent maneuvering strategy driven by knowledge and data collaboration was proposed. The method combines generative adversarial imitation learning with DDPG reinforcement learning algorithm, gave full play to the advantages of database and exploratory learning, thus ensured the convergence and robustness, and solved the problem of poor adaptability, long time-consuming and slow convergence in traditional penetration strategy optimization method.
(2) The initial position and velocity errors as well as the detection, guidance and control errors of the attack missile and the interceptor missile were fully established. Realistic constraints of the maneuver penetration were also considered in learning. A large number of adversarial simulation conditions were generated through Monte Carlo sampling, and Radau Pseudospectral method was used to search for the optimal solution offline.
(3) An incentive reward function that takes into account the target strike in the penetration process was designed to solve the problem of long-term sparse reward and accelerate the convergence of the algorithm to tackle problem of sparse reward in the reinforcement learning exploration process.

The manuscript is organized as follows. Section 2 describes the endo-atmospheric maneuver penetration problem of hypersonic vehicle and the relationship between optimal control and reinforcement learning. Research methods are detailed in Section 3. Simulation results are given in Section 4. Finally, Section 5 concludes the manuscript.

2. Problem formulation

The process of the penetration game between a single attack missile and an interceptor is shown as Fig. 1.

The single missile penetration adversarial game can be characterized by an OODA loop:

① Observe: A single missile obtains the interceptor's relative position, velocity, line of sight angle, and line of sight angular velocity information through the guidance head or the third outside guidance, forming a temporal detection information list.

② Orient: Attack missile conducts an estimation of penetration and confrontation capabilities, determines whether it is necessary to confront interceptors, and whether it has the ability to strike target after confronting with interceptor.

③ Decide: If confrontation is required, attack missile decides the optimal timing and command for maneuver confrontation. If confrontation is not required, attack missile flies according to the original planned guidance command.

④ Act: Attack missile maneuvers according to the timing and instructions of maneuver confrontation, completes endo-atmospheric penetration confrontation, and smoothly enter the mid to terminal guidance handover area.

According to the analysis of the penetration process, the missile maneuvering confrontation process is a typical Target Attack Defend (TAD) problem.[25] Thus, a relative motion relationship model is established, as shown in the following figure.

In Fig. 2, A represents the attack missile, D represents the interceptor missile, T represents the target which will be attacked by A. V and V_D are respectively the velocity of A and D. θ and θ_D are respectively the path angle of A and D. λ_{AD} is the line-of-sight angle between A and D.R_{AD} is the distance between A and D.λ_{AT} is the line-of-sight angle between A and T.R_{AT} is the distance between A and T.

The attack missile needs to comprehensively consider the constraints of penetration and target attack and perform optimal guidance calculation based on its own ballistic characteristics, which is a typical optimal control problem. To focus on penetration problem in gliding phase, the target accuracy constraint is equivalent to the accuracy and velocity constraints at the moment of mid to terminal guidance handover. Under those constraints and considering the attack missile's ability, the optimal penetration guidance command is solved to ensure the target can be successfully hit after penetration.

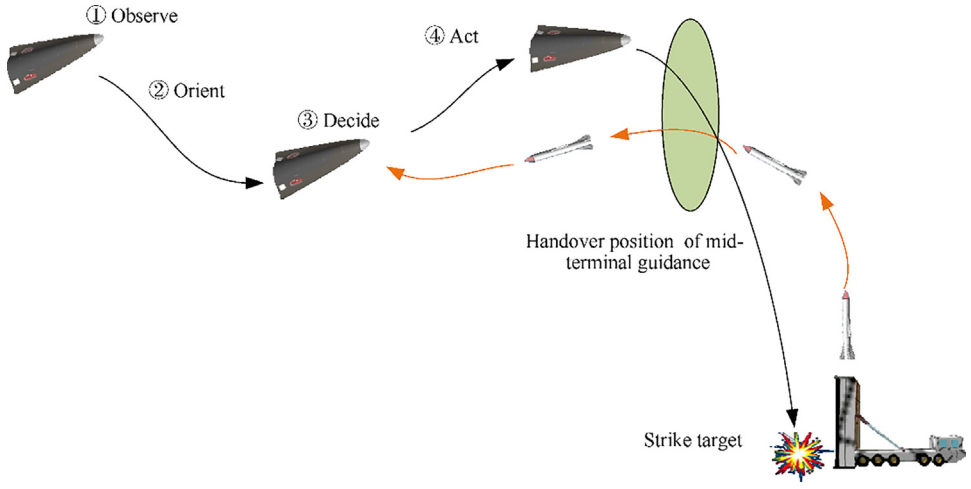


Fig. 1 Single missile penetration adversarial game process.

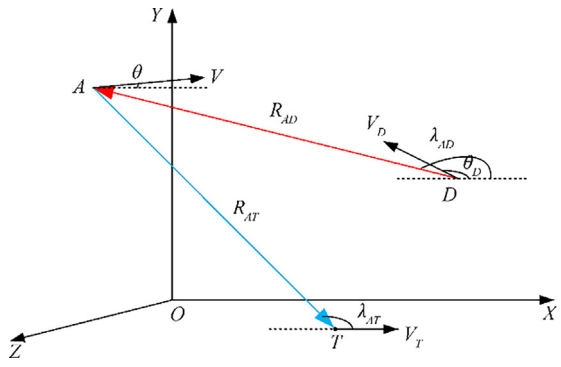


Fig. 2 Three body adversarial geometry.

2.1. Hypersonic vehicle model

The Hypersonic Glide Vehicle (HGV) is selected as the research subject, which has a maximum lift to drag ratio of 3.5, maximum speed 13 Ma, range of gliding phase about 800–1000 km.

For the convenience of designing penetration strategies, the aerodynamic coefficients of HGV are fitted using polynomial function. Taking into account the effects of AOA and Mach on lift and drag coefficients, the fitted model is as follows.

$$C_L = l_0 + l_1\alpha + l_2\alpha^2 + l_3 Ma + l_4 e^{l_5 Ma} \tag{1}$$

$$C_D = d_0 + d_1\alpha + d_2\alpha^2 + d_3 Ma + d_4 e^{d_5 Ma} \tag{2}$$

where α is Angle of Attack(AOA), Ma is the Mach number, C_L and C_D are the lift coefficient and drag coefficient, while l and d are coefficients. In this manuscript. We use a multivariate nonlinear least squares parameter identification method[26] to estimate the parameters in the above model. The fitted coefficients are as Tables 1 and 2.

The differential equations in the vehicle gliding phase can be formulated as follows:

$$\dot{\boldsymbol{x}} = \boldsymbol{f}[\boldsymbol{x}(t), \boldsymbol{u}(t), t] \tag{3}$$

where $\boldsymbol{x} = [x, y, z, V, \theta, \psi]$. $\boldsymbol{f}$ is dynamic and kinematic equation. x, y, z are respectively the components of the instantaneous position of the HGV on each axis of the ground coordinate system, ψ is deflection angle. $\boldsymbol{u} = [\alpha, \sigma]$ is control AOA and inclination angle. The detailed differential equations are listed in Eq. (4) with lift and drag equations in Eq. (5):

$$\begin{cases} \dot{x} = V\cos\theta\cos\psi \\ \dot{h} = V\sin\theta \\ \dot{z} = V\cos\theta\sin\psi \\ \dot{V} = -\frac{D}{m} - m\sin\theta \\ \dot{\theta} = \frac{L - mg\cos\theta}{mV} \\ \dot{\psi} = -\frac{Z}{mV\cos\theta} \end{cases} \tag{4}$$

$$\begin{cases} L = \frac{1}{2}\rho V^2 S C_L \\ D = \frac{1}{2}\rho V^2 S C_D \end{cases} \tag{5}$$

where L and D represents lift and drag, Z represents lateral force, m is the mass,S is the HGV reference area and ρ represents atmospheric density. Lateral plane motion is not considered in this manuscript.

Table 1 Fitted lift coefficients.

Lift coefficient	Value
l_0	–0.030 1
l_1	2.299 2
l_2	1.228 7
l_3	$-1.300\ 1 \times 10^{-5}$
l_4	0.204 7
l_5	$-6.114\ 60 \times 10^{-4}$

Table 2 Fitted drag coefficients.

Lift coefficient	Value
d_0	0.010 0
d_1	–0.174 8
d_2	2.724 7
d_3	$4.578\ 1 \times 10^{-6}$
d_4	0.359 1
d_5	$-6.944\ 0 \times 10^{-4}$

2.2. Interceptor missile model

According to reference 20, the Interceptor lift and drag coefficients can be listed as follows:

$$C_{LD} = (0.18 + 0.02 \cdot Ma_D) \cdot \alpha_D \tag{6}$$

$$C_{DD} = 0.18 + 0.01 \cdot Ma_D + 0.001 \cdot Ma_D \cdot \alpha_D + 0.004 \cdot \alpha_D^2 \tag{7}$$

where C_{LD} and C_{DD} are Interceptor lift and drag coefficient, Ma_D is the Interceptor Mach number, and α_D is the Interceptor AOA.

The movement equations of the interceptor can be listed as follows:

$$\begin{cases} \dot{x}_D = V_D\cos\theta_D\cos\psi_D \\ \dot{y}_D = V_D\sin\theta_D \\ \dot{z}_D = -V_D\cos\theta_D\sin\psi_D \\ \dot{V}_D = \frac{P\cos\alpha_D\cos\beta_D - D_D}{m_D} - m_D\sin\theta_D \\ \dot{\theta}_D = \frac{P\sin\alpha_D + L_D - m_D g\cos\theta_D}{m_D V_D} \\ \dot{\psi}_D = \frac{P\cos\alpha_D\sin\beta_D - Z_D}{m_D V_D\cos\theta_D} \end{cases} \tag{8}$$

where x_D, y_D, z_D are respectively the components of the instantaneous position of the interceptor on each axis of the ground coordinate system, ψ_D is deflection angle of the interceptor. L_D,D_D and Z_D are the lift, drag, and lateral force of the interceptor, m_D is the mass of the interceptor, and β_D is the sideslip angle of the interceptor.

The motion of the interceptor missile is guided by proportional guidance, satisfying

$$\dot{\theta}_D = k_D[V_D\sin(\lambda_{AD} - \theta_D) - V\sin(\lambda_{AD} - \theta)]/R_{AD} \tag{9}$$

where

$$R_{AD} = \| \boldsymbol{s}_A - \boldsymbol{s}_D \|_2 \tag{10}$$

In Eqs. (9) and (10), s_A and s_D are the positions of the attack missile and interceptor during the adversarial confrontation process respectively, and k_D is the proportional guidance coefficient.

2.3. Problem of optimization for penetration and striking

(1) Initial and terminal state constraints

The initial state of the attack missile is the flight state when the interceptor missile is about to be detected, and the initial state of the interceptor missile is the flight state close to the final guidance startup. Taking one of the typical states as an example, the initial state constraints are determined as follows:

$$\begin{cases} \boldsymbol{x}_{A0} = \boldsymbol{x}_{A0}^* \\ \boldsymbol{x}_{D0} = \boldsymbol{x}_{D0}^* \end{cases} \tag{11}$$

where $\boldsymbol{x}_{A0}$ is the initial state of the attack missile, $\boldsymbol{x}_{A0}^*$ is the value of the initial state of the attack missile, $\boldsymbol{x}_{D0}$ is the initial state of the interceptor, $\boldsymbol{x}_{D0}^*$ is the value of the initial state of the interceptor.

The attack missile needs to provide optimal initial conditions for mid to terminal guidance handover, and certain position and velocity constraints are also required in the gliding phase after penetration, to ensure meet terminal guidance conditions and ballistic energy requirement in the strike.

End state constraints are listed as follows:

$$\| \boldsymbol{s}_{Af} - \boldsymbol{s}_{Af}^* \|_2 \leqslant d \tag{12}$$

$$V_{Af} \geqslant V_{Af}^* \tag{13}$$

where $\boldsymbol{s}_{Af} = [x_{Af}, y_{Af}, z_{Af}]$, $\boldsymbol{s}_{Af}^* = \left[x_{Af}^*, y_{Af}^*, z_{Af}^*\right]$. $\boldsymbol{s}_{Af}$ and $\boldsymbol{s}_{Af}^*$ are respectively the position and expected position of the attack missile at the end of its interception penetration and into final guidance, with d is the maximum position deviation between them. $[x_{Af}, y_{Af}, z_{Af}]$ and $\left[x_{Af}^*, y_{Af}^*, z_{Af}^*\right]$ are respectively the three-dimensional position coordinates and expected position coordinates of the attack missile at the end of its interception penetration and into final guidance.

(2) Path constraints

Due to the harsh flight environment during the gliding phase of hypersonic aircraft, constraints on aerodynamic heating, overload and dynamic pressure are required. Thus, the penetration trajectory needs to meet the maximum overload, heat flux, and dynamic pressure constraints. The specific description is as follows:

$$\begin{cases} \| \boldsymbol{n} \|_2 \leqslant n_{\max} \\ q_A = 0.5\rho V_A^2 \leqslant q_{\max} \\ \dot{Q}_A = k_Q \rho^{0.5} V_A^{3.15} \leqslant \dot{Q}_{\max} \end{cases} \tag{14}$$

where $\boldsymbol{n}$ and $n_{\max}$ is respectively the actual overload and maximum overload constraint, q_A and $q_{\max}$ are respectively the actual dynamic pressure and maximum dynamic pressure constraint, $\dot{Q}_A$ and $\dot{Q}_{\max}$ is respectively the actual heat flux and maximum heat flux constraint, k_Q is heat flow constant. The maximum values of overload, dynamic pressure, and heat flux are 30 500 kPa, 6000 kW/m^2, respectively.

(3) Objective function

Precision strike is the main task of hypersonic vehicle, but to achieve such task, it is firstly necessary to ensure survival ability, which is to counter threats, and gain penetration advantages. Therefore, the optimization performance index for hypersonic vehicle trajectory requires comprehensive consideration of penetration efficiency and strike constraints, which determines the ability and characteristics of the optimal trajectory.

① Penetration indicators:

During the penetration process, the attack missile maximizes the distance between attack missile and the interceptor through ballistic maneuvers.

$$J_1 = -R_{AD} \tag{15}$$

② Precision strike indicators:

Target strike needs to be considered in the penetration process. Firstly, the position deviation between actual position and expected position entering the terminal guidance needs to be minimized. Furthermore, the velocity at the time of entering the terminal guidance needs to be maximized to retain energy for the terminal strike. Thus, precise strike indicators consists of two parts:

$$J_{21} = \| s_{Af} - s_{Af}^* \|_2 \tag{16}$$

$$J_{22} = -V \tag{17}$$

$$J_2 = J_{21} + J_{22} \tag{18}$$

③ Comprehensive performance indicators:

$$J = k_1 J_1 + k_2 J_2 \tag{19}$$

where J is the total performance index, k_1 and k_2 are weight coefficients.

2.4. Mathematical foundation of reinforcement learning solution to optimal control problem

Based on the constraints of penetration and strike for attack missile, the optimal control model equation is established with the optimization goal of successful penetration, achieving the highest accuracy and speed of mid to terminal guidance handover:

$$\begin{aligned} &\min J = \Phi[\boldsymbol{x}(t_{\mathrm{f}}), t_{\mathrm{f}}] \\ &\text{s.t. } \dot{\boldsymbol{x}}(t) = \boldsymbol{f}[\boldsymbol{x}(t), \boldsymbol{u}(t), t] \\ &C[\boldsymbol{x}(t), \boldsymbol{u}(t), t] \leqslant 0 \end{aligned} \tag{20}$$

where Φ is the endpoint performance indicator, t_0 and t_{f} are respectively the starting and ending times, respectively. C are the process constraints, including overload, stagnation point heat, and dynamic pressure constraints, $\boldsymbol{x}$ and $\boldsymbol{u}$ are defined above.

Discretize the motion equation of the vehicle

$$x(k+1) = \boldsymbol{f}[x(k), u(k)] \tag{21}$$

The optimization process is to obtain the optimal control variable at each step size

$$\boldsymbol{u}^* = [u^*(1), \cdots, u^*(k), \cdots, u^*(N)] \tag{22}$$

which is to achieve optimal performance index while meeting process constraints.

Obviously, each state transition in the optimization process satisfies

$$\boldsymbol{P}[x(k+1)|x(k)] = \boldsymbol{P}[x(k+1)|x(1), \cdots, x(k)] \tag{23}$$

Therefore, the optimization process is a Markov decision process and has Markovian characteristics.

The Markov decision process can be represented by a five tuple $\langle \mathscr{S}, \mathscr{A}, \mathscr{T}, \mathscr{R}, \gamma \rangle$, where $\mathscr{S}$ is a multidimensional state, $\mathscr{A}$ is an action space, $\mathscr{T}$ is a state transition function, satisfies $\mathscr{S} \times \mathscr{A} \rightarrow \mathscr{S}$, $\mathscr{R}$ is the timely reward obtained during the state transition process, γ is a discount factor, used to balance the relationship between timely reward and long-term reward.

The cumulative reward for each adversarial confrontation process can be recorded as:

$$G(s_0, \boldsymbol{u}) = \sum_{t=0}^{\infty} \gamma^t r_t \tag{24}$$

Under the optimal control instruction sequence $\boldsymbol{u}*$, the maximum cumulative reward can be expressed as

$$G^*(s_k) = \max_{a_t} \{R(s_k, a_k, s_{k+1}) + \gamma G^*(s_{k+1})\} \tag{25}$$

In the process of adversarial confrontation, the interceptor missile adopts proportional guidance to pursue the attack missile. The attack missile needs to make penetration instructions based on the state information of the interceptor missile. The adversarial confrontation environment is changing and unknown, thus the optimal problem of adversarial confrontation cannot be solved through simple dynamic programming and strategy iteration is required to learn the dynamic optimal strategy.

In the Markov decision process, each state value can be represented by a state value function $v_{\boldsymbol{\pi}}(\boldsymbol{s})$, and the value of an action in that state can be represented by an action value function $q_{\boldsymbol{\pi}}(\boldsymbol{s}, \boldsymbol{a})$. The state value function and action value function satisfy

$$\begin{aligned} v_\pi(\boldsymbol{s}) &= \textstyle\sum_{\boldsymbol{a} \in A} \pi(\boldsymbol{a}|\boldsymbol{s}) q_\pi(\boldsymbol{s}, \boldsymbol{a}) \\ &= \textstyle\sum_{\boldsymbol{a} \in A} \pi(\boldsymbol{a}|\boldsymbol{s}) (R_{\boldsymbol{s}}^{\boldsymbol{a}} + \gamma \sum_{\boldsymbol{s}' \in S} P_{SS}^{a}, v_\pi(\boldsymbol{s}')) \end{aligned} \tag{26}$$

$$\begin{aligned} q_\pi(\boldsymbol{s}, \boldsymbol{a}) &= R_{\boldsymbol{s}}^{\boldsymbol{a}} + \gamma \textstyle\sum_{\boldsymbol{s}' \in S} P_{SS}^{a}, v_\pi(\boldsymbol{s}') \\ &= R_{\boldsymbol{s}}^{\boldsymbol{a}} + \gamma \textstyle\sum_{\boldsymbol{s}' \in S} P_{SS}^{a}, \sum_{\boldsymbol{a}' \in A} \pi(\boldsymbol{a}'|\boldsymbol{s}') q_\pi(\boldsymbol{s}', \boldsymbol{a}') \end{aligned} \tag{27}$$

Using strategy iteration method, the action with the highest value is selected each time as the new strategy, new strategy is updated to recalculate the state and action values until convergence condition is reached.

3. GAIL-DDPG endo-atmospheric maneuver penetration strategy

In this section, a novel GAIL-DDPG Endo-atmospheric maneuver penetration strategy is detailed. Firstly, the optimal control problem is transformed into a nonlinear programming problem utilizing Radau Pseudospectral method to discretize the optimal control problem at the selected orthogonal mating point, according to the characteristics of the optimal control problem in the process of penetrating and attacking. Radau Pseudospectral method is used to solve the transformed problem to generate the data sample of the penetration attack trajectory planning considering initial conditional bias and error disturbance. Then, the GAIL-DDPG algorithm network is constructed with GAIL network trained based on generated data samples and performance further optimized by DDPG, as shown in Fig. 3. The proposed method is then further verified with simulation comparison with non-maneuvering, random maneuvering, and DDPG strategies.

3.1. Radau Pseudospectral solution to optimal control problem

The Radau Pseudospectral method is a collocation method for solving multi-constraint optimal control problems, which has advantages of good convergence and high solution accuracy. The method is to discretize the state and control on a series of orthogonal nodes, namely Legendre Gauss Radau (LGR) nodes, satisfying the state differential equations. The obtained discrete points are used as nodes to construct Lagrange interpolation polynomials to approximate the state and control variables, thereby transforming the optimal control problem into a parameter optimization problem.

Assuming the time interval for trajectory planning is $t \in [t_0, t_f]$, the continuous time domain is divided into K networks. For each single grid $[t_{k-1}, t_k]$, the basic steps for applying the Radau Pseudospectral method are as follows:

(1) Time-domain transformation

Convert the time interval to $\tau \in [-1, 1]$, perform transformation

$$\tau \in [2t - (t_k + t_{k-1})]/(t_k - t_{k-1}), t_{k-1} < t_k \tag{28}$$

$$\mathrm{d}\tau/\mathrm{d}t = 2/(t_k - t_{k-1}); k = 1, 2 \cdots K \tag{29}$$

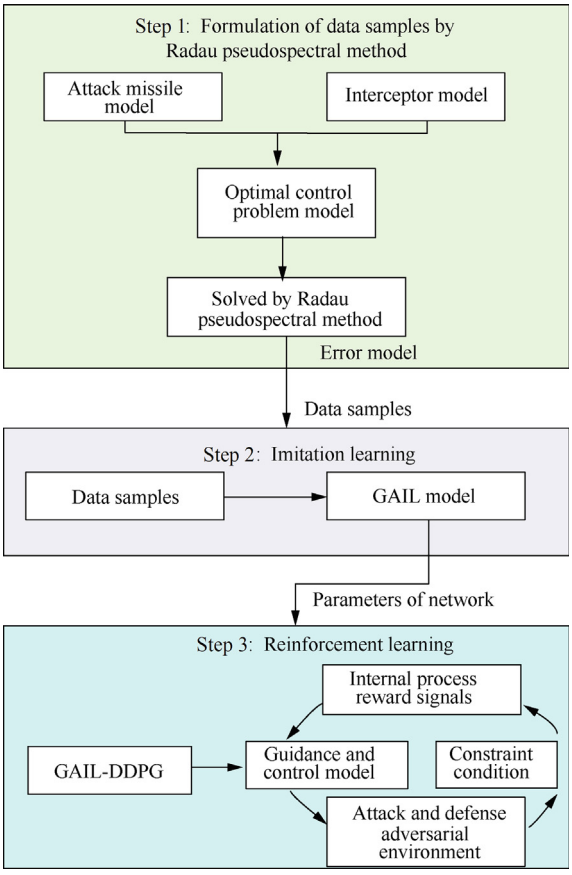


Fig. 3 GAIL-DDPG Endo-atmospheric maneuver penetration strategy.

(2) Discretization of control variables and state variables

Assuming that there are a total of N_k collocation points in the k th network, i.e.$\left\{\tau_1^{(k)}, \tau_2^{(k)}, \cdots, \tau_{N_k}^{(k)}\right\}$, approximating the state variables $x^{(k)}(\tau)$ in the k th grid through Lagrange interpolation polynomials

$$x^{(k)}(\tau) \approx \sum_{j=1}^{N_k+1} X^{(k)}\left(\tau_j\right) L_j^{(k)}(\tau) \tag{30}$$

$$L_j^{(k)}(\tau) = \prod_{\substack{l=1 \\ l \neq j}}^{N_k+1} \frac{\tau - \tau_l^{(k)}}{\tau_j^{(k)} - \tau_l^{(k)}};\ j = 1, 2, \cdots, N_k + 1 \tag{31}$$

where $\tau \in [-1, 1]$, $L_j^{(k)}(\tau)$ represents the Lagrange interpolation basis function of order N_k. The approximation is done through N_k order Lagrange polynomials with control variable $u^k(\tau)$

$$u^k(\tau) \approx \sum_{j=1}^{N_k+1} U^{(k)}\left(\tau_j\right) \widehat{L}_j^{(k)}(\tau) \tag{32}$$

$$\widehat{L}_j^{(k)}(\tau) = \prod_{\substack{i=1 \\ i \neq j}}^{N_k+1} \frac{\tau - \tau_i^{(k)}}{\tau_j^k - \tau_i^{(k)}} \tag{33}$$

where $k = K, \tau \in \left[\tau_1^{(k)}, \tau_{N_k+1}^{(k)}\right]$.

(3) Discretization of constraint conditions

Take derivative of $x^{(k)}(\tau)$:

$$\frac{\mathrm{d}x^{(k)}(\tau)}{\mathrm{d}\tau} = \dot{x}^{(k)}(\tau) \approx \sum_{j=1}^{N_k+1} X^{(k)}\left(\tau_j\right) \dot{L}_j^{(k)}(\tau) \tag{34}$$

Substitute Eq. (24) into the optimal control equation and discretize at each configuration point to obtain

$$\sum_{j=1}^{N_k+1} X^{(k)}\left(\tau_j^{(k)}\right) D_{ij}^{(k)} - \frac{t_k - t_{k-1}}{2} f\left(X^{(k)}, U^{(k)}, \tau_j^{(k)}, t_{k-1}, t_k\right) = 0 \tag{35}$$

where $D_{ij}^{(k)} = \dot{L}_j^{(k)}\left(\tau_j^{(k)}\right);\ i = 1, 2, \cdots, N_k; j = 1, 2, \cdots, N_k + 1$.

Discretize the boundary constraints and inequality constraints at the selected configuration points to obtain

$$\phi\left(x^{(k)}(-1), t_{k-1}, x^{(k)}(+1), t_k\right) = 0 \tag{36}$$

$$C\left(x^{(k)}(\tau), u^{(k)}(\tau), \tau_i^{(k)}; t_{k-1}, t_k\right) \leqslant 0 \tag{37}$$

To ensure that the selected grid points are continuous, it is necessary to ensure

$$x^{(k)}(+1) = x^{(k+1)}(-1) \tag{38}$$

(4) Discretization of performance indicators

Discretize the performance index function and approximate it as follows:

$$J \approx \Phi\left(X_1^{(1)}, t_0, X_{N_k+1}^K, t_K\right) \tag{39}$$

By applying the aforementioned steps, the optimal control problem is transformed into a nonlinear control problem.

Radau Pseudospectral method are utilized to directly solve the motion states of both attack and interception sides, avoiding numerical integration of differential equations to solve interception trajectories, thus computational complexity is reduced. Select the relevant flight state variables of interception and attack missiles as the state variables.

$$\begin{cases} \boldsymbol{x}_A = [x, y, V, \theta, \alpha] \\ \boldsymbol{x}_D = [x_D, y_D, V_D, \theta_D, \alpha_D] \end{cases} \tag{40}$$

3.2. Data samples generation

Data generated by Radau Pseudospectral method is used as samples for GAIL-DDPG imitation learning, and the penetration trajectory library can be represented as

$$\boldsymbol{\chi} = \left\{\boldsymbol{\zeta}^{(1)}, \boldsymbol{\zeta}^{(2)}, \cdots, \boldsymbol{\zeta}^{(l)}\right\} \tag{41}$$

where each trajectory is $\boldsymbol{\zeta}^{(l)} = \left[\boldsymbol{s}_1^{(l)}, \boldsymbol{a}_1^{(l)} \boldsymbol{s}_2^{(l)}, \boldsymbol{a}_2^{(l)} \cdots, \boldsymbol{s}_n^{(l)}, \boldsymbol{a}_n^{(l)}\right]$.

The penetration trajectory library is constructed by considering random process errors. The random process is shown in Table 3, which includes the guidance head detection errors (or detection errors from the third party), missile attitude control delay, and initial position and velocity deviation errors of attack and interceptor missiles.

A total of 1000 simulations were performed, and the sample data generated is shown in the Fig. 4.

Table 3 Random errors in trajectory generation.

Parameter	HGV	Interceptor	Type of error
Detection distance (km)	40 ± 10	20 ± 2	Uniform distribution
Blind spot (m)	200–500	200–500	Uniform distribution
Angle measurement error	Zero mean	Zero mean	Gaussian distribution
Sight rotation rate measurement error	Zero mean	Zero mean	Gaussian distribution
Ranging error	Zero mean	Zero mean	Gaussian distribution
Velocity measurement error	Zero mean	Zero mean	Gaussian distribution
First order delay	0.1–1	0.1–1	Uniform distribution
Initial position deviation (km)	±3	±2	Uniform distribution
Initial velocity deviation (m/s)	±600	±300	Uniform distribution

3.3. Penetration strategy network based on GAIL-DDPG

In this manuscript, a novel penetration strategy network based on GAIL-DDPG is proposed, which combines the advantages of GAIL and DDPG algorithms. GAIL is a type of imitation learning that quickly improves the network's policy generation capabilities by letting the policy network learn from experts, and at the same time, DDPG is an actor-critic framework method suitable for continuous control problems, continues the experience replay mechanism in the DQN algorithm to minimize the correlation between samples. The policy network and the value network are separated, which makes the policy exploration and policy update are performed in a time-shared manner, making the training process more stable. The advantages of GAIL and DDPG are combined in the proposed method twofold. Firstly, it provides expert data samples through the micro-GAIL network to support rapid network convergence and shorten training time. Secondly it combines the DDPG algorithm for strategy exploration to further improve the generalization ability and optimization performance of the network.

3.3.1. Algorithm network construction based on GAIL-DDPG

The GAIL-DDPG method is shown as Fig. 5.

The GAIL-DDPG method consists of two parts. The first part of the network learns through GAIL network from the data samples obtained by the Radau Pseudospectral method to improve the network training speed. When the generator network converges or reaches the maximum number of training times N_{GAIL}, the network parameters are assigned to actor network. The latter part of the network is based on the DDPG algorithm under the Actor-Critic framework and is trained in a hot-start mode. The generalization ability of the network is further improved by adding exploration noise.

The generator's policy network can be expressed as:

$$\boldsymbol{g} = \pi(\cdot|\boldsymbol{s};\boldsymbol{\theta}) \tag{42}$$

where the dimension of the output vector $\boldsymbol{g}$ is the size of the missile maneuver action space $\mathscr{A}$, and each element corresponds to an action, representing the probability of executing that action. Given the initial state and allowing the agent to interact with the environment, a trajectory can be obtained:

$$\boldsymbol{\zeta} = [\boldsymbol{s}_1, \boldsymbol{a}_1, \boldsymbol{s}_2, \boldsymbol{a}_2, \cdots, \boldsymbol{s}_n, \boldsymbol{a}_n] \tag{43}$$

The discriminator of GAIL-DDPG is denoted as $D(\boldsymbol{s}, \boldsymbol{a}; \phi)$, with input $(\boldsymbol{s}, \boldsymbol{a})$ and output of a vector. Each element in the vector represents the discrimination result for the state action pair. The closer the discrimination result is to 1, the closer the generator strategy is to the expert strategy.

The actor network of GAIL-DDPG consists of an online policy network and a target policy network. The policy network consists of a fully connected neural network. The output action is determined with a certain state input. The target policy network updates variables through soft updates.

The critic network of GAIL-DDPG consists of an online value network and a target value network. The value network is also composed of a fully connected neural network, which is used to evaluate the state action value and reflect the quality of the action. During the training process, the value network helps train the policy network. After training, the value network is discarded and the policy network controls the agent. The target value network updates variables through soft updates.

(1) State space:

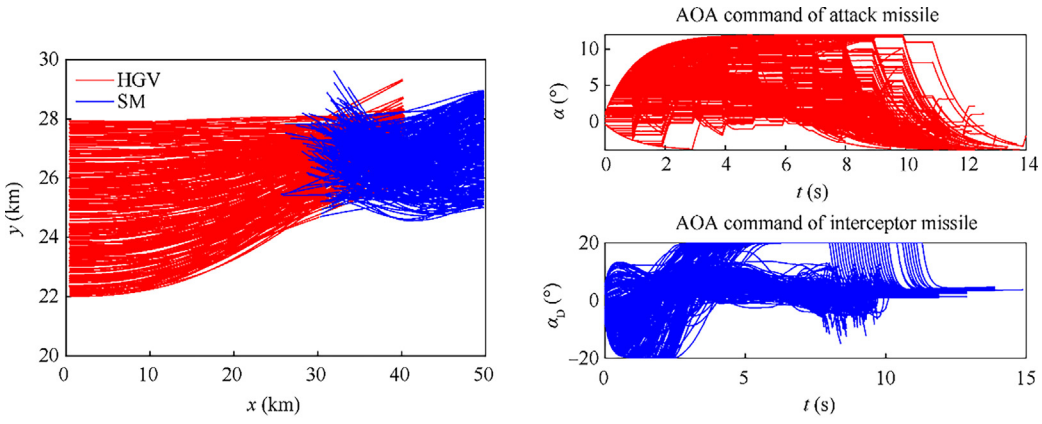
For endo-atmospheric penetration confrontation, based on the assumption that the attack missile carries an active radar detector, the attack missile can obtain the position and speed information of the interceptor missile when the interceptor enters the attack missile detection field of view. Therefore, the state space can be built as:

$$\boldsymbol{s} = \left[x, y, V, \theta, \alpha, n_{\max}, x^*_{\mathrm{Af}}, y^*_{\mathrm{Af}}, x_D, y_D, V_D\right] \tag{44}$$

The definition of variables can be found in preceding context.

(2) Action space:

The attack missile action is defined by its AOA. The penetration problem can be solved by GAIL-DDPG with a continuous action space, which describe realistic missile movement. The action space is then continuous with value range of [–4°,12°].



(a) Trajectory sample (b) Control commands

Fig. 4 Adversarial trajectory sample.

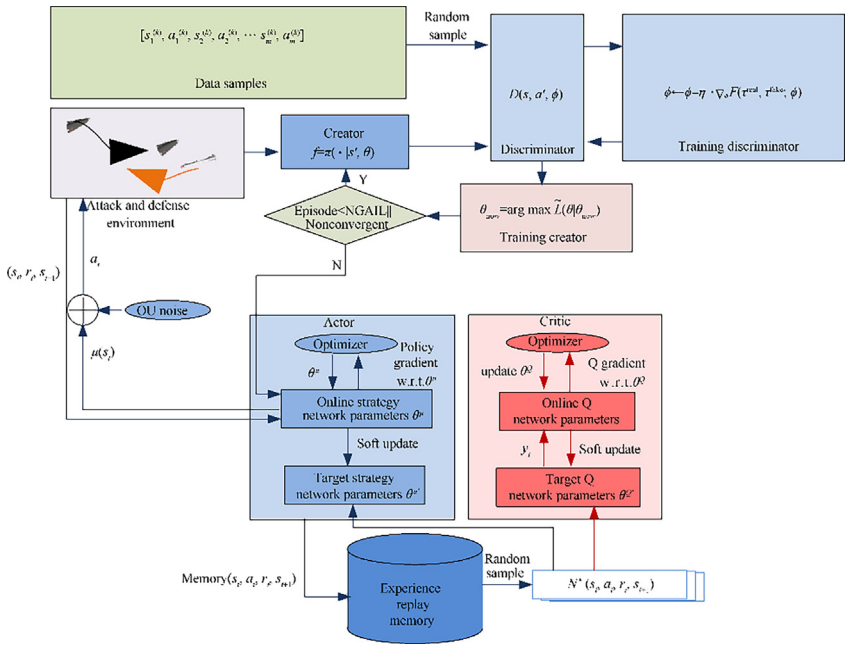


Fig. 5 GAIL-DDPG method.

(3) Reward function design:

The reward function consists of three parts. The first part is the reward for reaching the mid to terminal guidance handover area, the second part is the reward for the mid to terminal guidance handover velocity, and the third part is the reward for the penetration process. If the penetration is failed, the total reward will be zero. Thus, the reward function is as follows and is shown in Fig. 6.

$$\begin{cases} R_1 = \frac{5}{0.5+\|s_{Af}-s_{Af}^*\|_2} \\ R_2 = 0.003 \times (V_{Af} - 2\ 000) \\ R_3 = \log_{10}(R_{AD} - 149) \end{cases} \tag{45}$$

$$R = \begin{cases} 0, & \text{If the penetration is failed} \\ R_1 + R_2 + R_3, & \text{Otherwise} \end{cases} \tag{46}$$

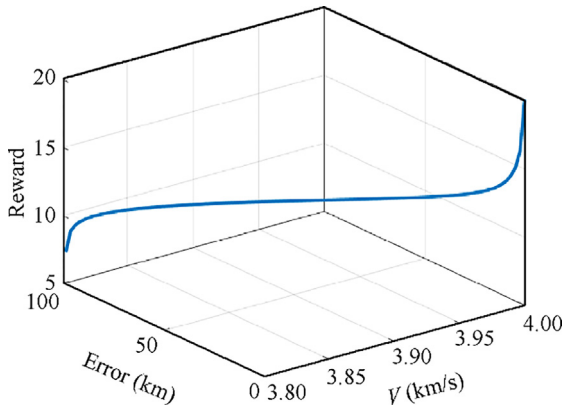


Fig. 6 Reward function.

3.3.2. GAIL-DDPG algorithm network training

The training steps for GAIL-DDPG network can be listed as follows:

Step 1. Uniformly sample a trajectory from the training set, denoted as

$$\boldsymbol{\zeta}^{\text{real}} = \left[\boldsymbol{s}_1^{\text{real}}, \boldsymbol{a}_1^{\text{real}}, \cdots, \boldsymbol{s}_n^{\text{real}}, \boldsymbol{a}_n^{\text{real}}\right] \tag{47}$$

Step 2. Using strategy network $\pi(\boldsymbol{a}|\boldsymbol{s};\boldsymbol{\theta}_{\text{now}})$ to control the interaction between intelligent agent and the environment, obtaining a trajectory, denoted as

$$\boldsymbol{\zeta}^{\text{fake}} = \left[\boldsymbol{s}_1^{\text{fake}}, \boldsymbol{a}_1^{\text{fake}}, \cdots, \boldsymbol{s}_n^{\text{fake}}, \boldsymbol{a}_n^{\text{fake}}\right] \tag{48}$$

Step 3. Using discriminator to evaluate the authenticity of decisions in policy networks

$$u_t = \ln D\left(\boldsymbol{s}_t^{\text{fake}}, \boldsymbol{a}_t^{\text{fake}}; \phi_{\text{now}}\right);\ \forall t = 1, \cdots, n \tag{49}$$

Step 4. Using $\boldsymbol{\zeta}^{\text{fake}}$ and $u_1, \cdots, u_n$ as inputs, update the generator network parameters by Trust Region Policy Optimization to obtain $\boldsymbol{\theta}_{\text{new}}$

$$\boldsymbol{\theta}_{\text{now}} = \arg\max_{\theta} \tilde{L}\left(\boldsymbol{\theta}|\boldsymbol{\theta}_{\text{now}}\right);\quad \text{s.t. } \text{dist}(\boldsymbol{\theta}_{\text{now}}, \boldsymbol{\theta}) \leqslant \Delta \tag{50}$$

where

$$\tilde{L}\left(\boldsymbol{\theta}|\boldsymbol{\theta}_{\text{now}}\right) = \frac{1}{n}\sum_{t=1}^{n} \frac{\pi(\boldsymbol{a}_t|\boldsymbol{s}_t;\boldsymbol{\theta})}{\pi(\boldsymbol{a}_t|\boldsymbol{s}_t;\boldsymbol{\theta}_{\text{now}})} \cdot u_t \tag{51}$$

Step 5. Using $\boldsymbol{\zeta}^{\text{real}}$ and $\boldsymbol{\zeta}^{\text{fake}}$ as input and update the discriminator network parameters by gradient descent method to update ϕ

$$\phi \leftarrow \phi - \eta \cdot \nabla_{\phi} F\left(\boldsymbol{\zeta}^{\text{real}}, \boldsymbol{\zeta}^{\text{fake}}; \phi\right) \tag{52}$$

where

$$F(\zeta^{\text{real}},\zeta^{\text{fake}};\phi)=\frac{1}{m}\sum_{t=1}^{m}\ln\left[1-D\left(\boldsymbol{s}_t^{\text{real}},\boldsymbol{a}_t^{\text{real}};\phi\right)\right]+\frac{1}{n}\sum_{t=1}^{n}\times\ln D\left(\boldsymbol{s}_t^{\text{fake}},\boldsymbol{a}_t^{\text{fake}};\phi\right) \tag{53}$$

Step 6. Repeat the above steps. When the network converges or the number of updates reaches N_{GAIL}, assign the generator network parameters to the Actor's Online strategy network.

Step 7. Initialize exploration noise, generate actions based on the current strategy and exploration noise, perform state transitions based on actions and observations, and store the transitions $(\boldsymbol{s}_t,\boldsymbol{a}_t,r,\boldsymbol{s}_{t+1})$ in the experience replay pool.

Step 8. After the number of storages is reached, N batches of samples are randomly sampled from the experience replay pool, and the critic network is updated based on the minimized loss function. The loss function is as follows

$$L=\frac{1}{N}\sum_{i}\left(y_i-Q(\boldsymbol{s}_i,\boldsymbol{a}_i|\theta^Q)\right)^2 \tag{54}$$

where $y_i=r_i+\gamma Q'\left(s_{i+1},\mu'\left(\boldsymbol{s}_{i+1}|\theta^{\mu'}\right)|\theta^{Q'}\right)$

Step 9. Updating actor network through gradient descent method

$$\nabla_{\theta^\mu}J\approx\frac{1}{N}\sum_{i}\nabla_a Q(\boldsymbol{s},\boldsymbol{a}|\theta^Q)|_{\boldsymbol{s}=\boldsymbol{s}_i,\boldsymbol{a}=\mu(\boldsymbol{s}_i)}\nabla_{\theta^\mu}\mu(\boldsymbol{s}|\theta^\mu)|_{\boldsymbol{s}_i} \tag{55}$$

Step 10. After a certain period, update the target network

$$\theta^{Q'}\leftarrow\xi\theta^Q+(1-\xi)\theta^{Q'} \tag{56}$$

$$\theta^{\mu'}\leftarrow\xi\theta^\mu+(1-\xi)\theta^{\mu'} \tag{57}$$

where ξ is the soft update parameter.

Step 11. Repeat Steps 7 to 10 until the network converges or reaches the maximum number of iterations, then stop the network training.

4. Simulation result

In this section, the effectiveness of proposed penetration strategy is validated through numerical simulations. The simulation conditions is shown in Section 4.1, with results analyzed in Section 4.2.

4.1. Simulation conditions

The simulations are conducted with a realistic attack missile and interceptor model, with the conditions shown in Table 4.

4.2. Simulation analysis

The simulation is conducted in training platform with processor AMD Ryzen5 3600@4.2 GHz and RAM 8 GB × 2 DDR4@3733 MHz. To compare the algorithm performance of the GAIL-DDPG network and the DDPG network, they were trained separately. The training results are shown in Fig. 7.

The DDPG algorithm converges after about 800 episodes, and the average reward is close to 19.5. The GAIL-DDPG algorithm converges after about 500 episodes, and the average reward is close to 20.3. It can be seen from the figure that the DDPG algorithm has been explored for a long time. After a period of exploration, the rewards begin to rise and the training process is slow. Because the GAIL-DDPG penetration strategy network firstly learns from expert experience, it can obtain better reward values faster. The Reward curve oscillates around 11, and switches to the DDPG network at about episode 200. The Reward is further increased after strategy exploration. After training for 400 episodes, the network performance is basically close to the pure DDPG algorithm, and the training speed is increased by 37.5%. After training for 600 episodes, the network performance is greatly improved, and the final average reward is increased by 2.8%.

Table 4 Vehicle parameters.

Parameter	Interceptor	Attack missile
Mass (kg)	75	500
Reference area (m^2)	0.3	0.579
Minimum speed (m/s)	400	1000
AOA (°)	–20–20	–4–12
typical Detection distance (km)	20	50
Detector off-axis angle (°)	45	10
Attitude control time constant	0.1	1
Initial position x (km)	50	0
Initial position y (km)	27	25
Initial velocity (m/s)	2000	4000
Initial ballistic inclination angle (°)	0	0
Center of mid-terminal handover x (km)		40
Center of mid-terminal handover y (km)		28
Interceptor killing radius (m)	150	

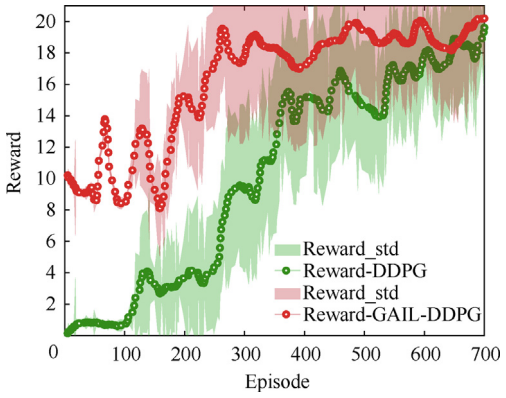


Fig. 7 Training reward of GAIL-DDPG.

To compare the performance, 1000 Monte Carlo simulations were conducted for the GAIL-DDPG, DDPG, the fixed trajectory and random maneuver penetration strategy. The comparison table is listed in Table 5. Attack missile is difficult to successfully penetrate without maneuvering or with random maneuvers. The success rates of penetration and being able to accurately reach the mid-to-terminal guidance shift center point are extremely small. Both the DDPG algorithm and the GAIL-DDPG algorithm can complete the penetration and strike mission. The DDPG algorithm takes longer to train while the GAIL-DDPG algorithm can explore more fully and have better performance. In contrast, the GAIL-DDPG algo-

Table 5 Comparison of algorithm performance.

No.	Category	Penetration success rate (missing distance 300 m) (%)	Target reaching error (less than 1 km) (%)	Remaining velocity (greater than 2000 m/s) (%)	All conditions meet (%)
1	No maneuver	0.7	0.1	100	0.1
2	random maneuver	1.7	0	100	0
3	DDPG	94.6	91.4	92.3	79.8
4	GAIL-DDPG	96.2	94.1	94.5	83.3

rithm has satisfactory performance for the training time is shorter and the engineering applicability is better.

To further demonstrate the performance, two intelligent penetration simulation results are detailed in sections below.

(1) Horizontal flight and suddenly jump for penetration and striking

It can be seen from Fig. 8 and Fig. 9 that after the interceptor missile enters the attack missile's field of view, the attack missile adopts a jump maneuver by using variable overload, and the ballistic trajectory is flat then pulls up rapidly. Fig. 10 shows that because the initial position of the interceptor missile is high above and adopts rapid dive proportional guidance to attack the target, after the trajectory of the attack missile pulls up, the interceptor missile begins to pull up its trajectory with a large overload maneuver, resulting in severe saturation of the maneuver and eventually misses the attack missile. As can be seen from Fig. 11, variable AOA is utilized by the attack missile, and is adjusted according to the requirement of penetration and strike mission. While the overload of Interceptor saturates between 1–3 s and 4–9.3 s.

It can be seen from Fig. 12 that the terminal velocity of the attack missile is 3500 m/s, and a large portion of kinetic energy is saved before entering terminal guidance. Fig. 13 shows that the interceptor missile's final miss distance is about 480 m.

(2) Jumping down to deceive and then pull up for penetration and striking

Another typical maneuver penetration strategy learned by proposed method can be summarized as firstly perform deceive

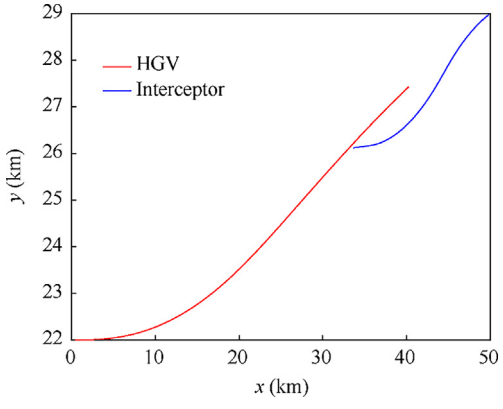


Fig. 8 Trajectories of Sample 1.

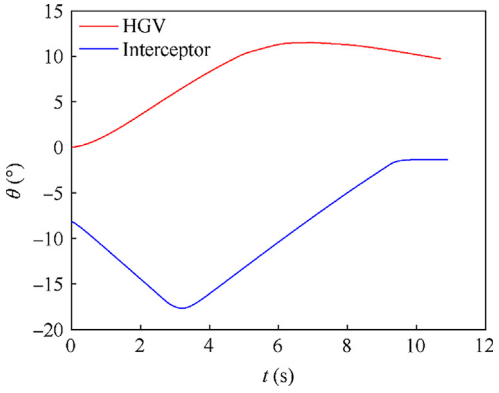


Fig. 9 Trajectory inclinations of Sample 1.

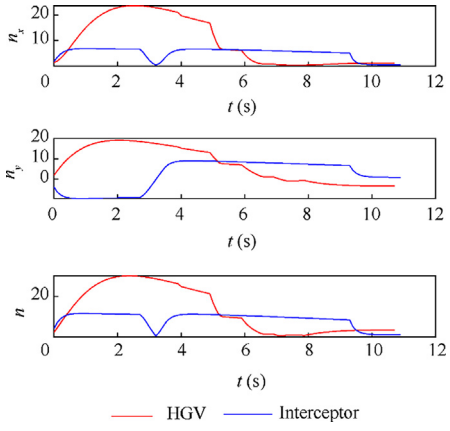


Fig. 10 Three-dimensional accelerations of Sample 1.

move, and then act an opposite maneuver to penetrate. The results are shown in Figs. 14–19. Fig. 14 and Fig. 15 show that the attack missile is in an initial disadvantage confrontation airspace when close to level flight. A downward jump decoy maneuver is firstly performed to lower height. After interceptor follows the move, the attack missile suddenly pulls up and fly towards the guidance handover point. The interceptor missile loses energy due to the deceive maneuver, and overload saturation in the latter move results in the loss of the target.

As can be seen from Fig. 16 and Fig. 17, the attack missile adopts a long-term large load maneuver, and the interceptor missile's acceleration changes accordingly. The interceptor

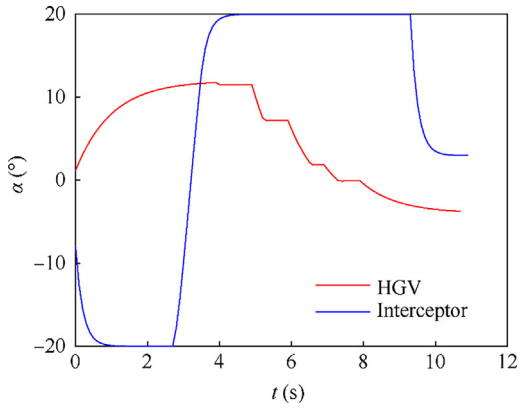


Fig. 11 AOAs of Sample 1.

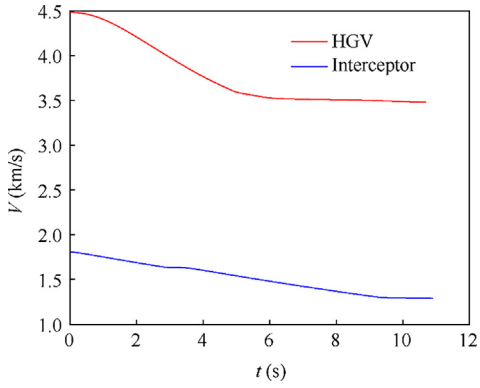


Fig. 12 Velocities of Sample 1.

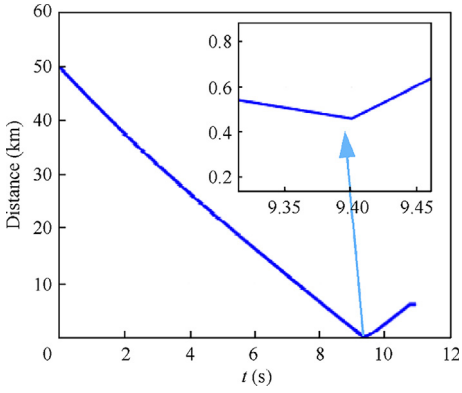


Fig. 13 Distance of Sample 1.

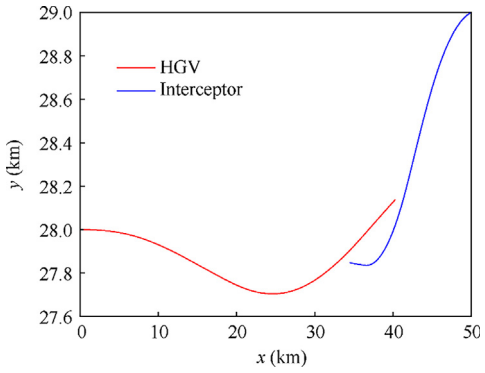


Fig. 14 Trajectories of Sample 2.

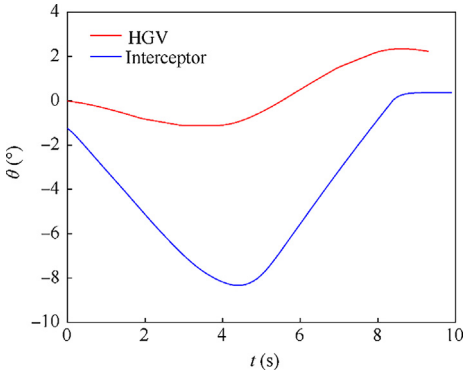


Fig. 15 Trajectory inclinations of Sample 2.

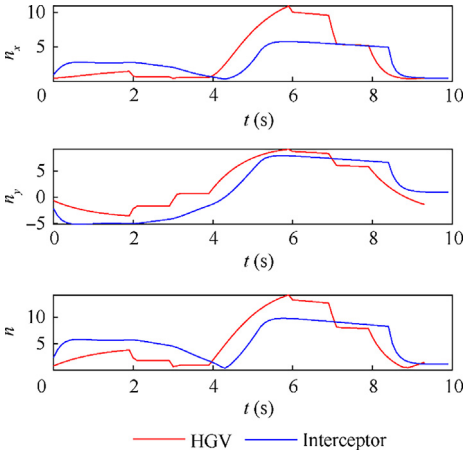


Fig. 16 Three-dimensional accelerations of Sample 2.

missile's overload saturates between 8 s and 8.5 s due to the limitation of the AOA, resulting in the final miss. Fig. 18 shows the velocity of the attack missile decreases slightly after the initial altitude decreases, and the velocity decreases to 4 100 m/s as the trajectory is pulled up. The velocity of the interceptor missile decreases from 1 700 m/s to 1 400 m/s, and the altitude drop compensates for part of the velocity loss. It can be seen from Fig. 19 that the final miss distance is 200 m.

5. Conclusions

An intelligent penetration maneuvering method for attack missile based on a generative adversarial reinforcement learning is proposed in this manuscript to solve the problem of *endo*-atmospheric maneuver penetration of gliding vehicle. Numerical simulation has been conducted to analyze the performance of the proposed method.

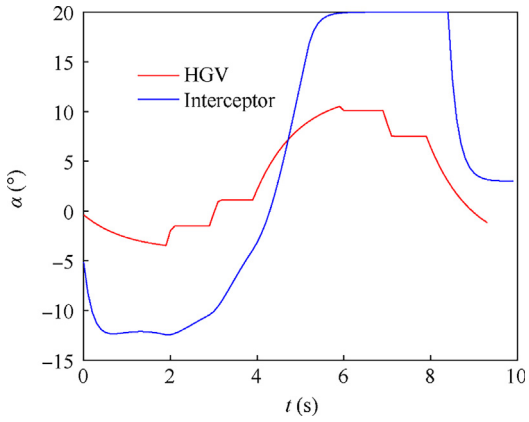


Fig. 17 AOAs of Sample 2.

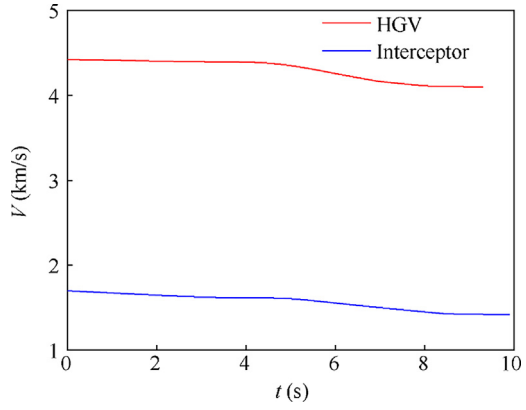


Fig. 18 Velocities of Sample 2.

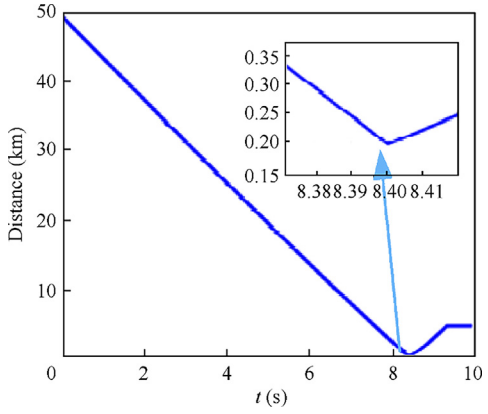


Fig. 19 Distance of Sample 2.

(1) Compared with the traditional reinforcement learning algorithm network, the GAIL-DDPG proposed in this manuscript provides expert data samples through micro-GAIL network to support rapid network convergence, strategy exploration is also taken advantage to further improve with shorter training time and greater performance.

(2) Process reward function has been designed to improve the convergence speed to tackle long-term sparse reward in the penetration confrontation process, and attack missile can save energy as much as possible while satisfy the requirement of penetration and the accuracy of mid-to-terminal guidance handover.

(3) The proposed GAIL-DDPG method can adapt to various types of penetration and confrontation situations. In non-dominant penetration situations, it can still flexibly exert its maneuverability and skillfully cope with interception and has certain intelligence emergence capability.

CRediT authorship contribution statement

Yaoluo HUI: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xiumin LI:** Supervision, Conceptualization. **Chen LIANG:** Writing – review & editing. **Junzheng SUN:** Methodology. **Zheng DU:** Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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